

Projects for CS 4366 – Fall 2026

1. Chat with Github: Enhanced GitHub Repository Analysis with Large Language Models

Navigating through a GitHub repository to comprehend or modify its functions can be daunting, especially for programming beginners. Typically, one might need to delve into every function and perhaps even run it locally, dealing with environment setups, which can be quite time-consuming. Although one can paste code snippets from GitHub to ChatGPT for explanations, several challenges arise:

- **Code Distribution:** A GitHub repository often spreads code over multiple files and directories, making it cumbersome to ask questions about the entire codebase.
- **Model Memory Limit:** If a code snippet is extensive, models like ChatGPT may forget earlier parts of the code, leading to incomplete or inaccurate responses.
- **Overlooking Issues/Discussion Content:** Developers often provide valuable insights in the Issues or Discussion section, which can easily be missed.

To address this, we propose leveraging large language models to dissect GitHub repositories into manageable chunks, converting them into a vector database. When a query is posed, the model can fetch relevant code snippets and provide explanations. **Additionally, it can explain the answers via interactive diagrams or visualizations that allow the users to navigate and understand how the answers were generated.** Projects like chat-pdf on GitHub, which embed and match vectors to explain PDF content, can serve as a blueprint. Explore similar work here: <https://www.chatpdf.com/>.

2. Reengineering the project specification from its source code

Reengineering a project specification from its source code involves reverse engineering, which is the process of analyzing and understanding the existing codebase to extract a high-level specification or design documentation. This reengineering task first needs to conduct a thorough review of the source code to understand its structure, logic, and components, and identify and document the various modules, functions, and classes present in the code. Then, it will further extract information from code comments that provide insights into the purpose and functionality of different code segments. In particular, the teams will be required to generate design diagrams, such as sequence diagrams or class diagram of example projects.

3. Code generation from flowcharts

A flowchart is a type of diagram that represents an algorithm, workflow or process. The flowchart shows the steps as boxes of various kinds, and their order by connecting the boxes with arrows. Code generation from flowcharts involves translating the diagrammatic representation of a program (flowchart) into executable code. It is essential to note that fully automated code generation from flowcharts is a complex task. It needs to map flowchart elements (symbols) to corresponding code constructs. For example, rectangles might represent procedural code blocks, diamonds might represent conditional statements (if-else), and arrows might represent the flow of control between code blocks. Besides, it also needs to handle the loop structures and error handlings in the flowchart. **The flowcharts and the code needs to be linked on the generated interface.**

4. Reengineering the flowchart from a source code

Reengineering a flowchart from existing source code involves analyzing the code to extract the logical structure and control flow, then representing that information visually in a flowchart. It needs to understand the source code structure, including functions, loops, conditionals, and other control flow structures, and identify the main components, such as functions or methods, and their relationships. Each function may correspond to a distinct block in the flowchart. It also needs to trace the control flow through the code, noting how decisions (if statements) and loops are structured, to identify branching points and how the program responds to different conditions, and to identify sections of code dedicated to error handling or exception handling and loop structures. These may be represented as decision points in the flowchart.

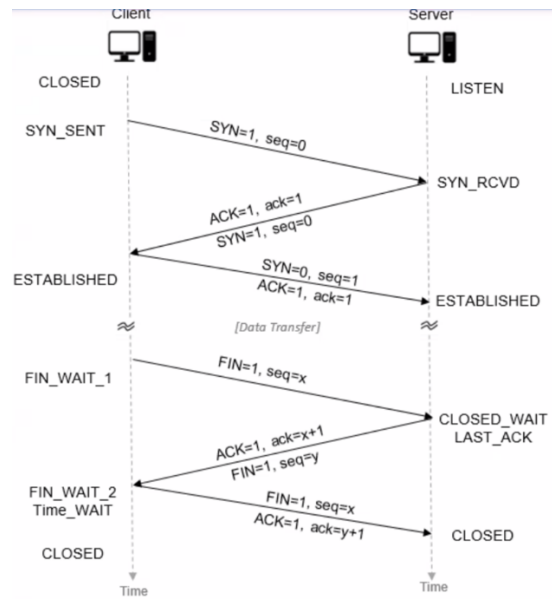
And finally, it will create a visual representation of the code by drawing a flowchart using standard symbols for processes, decisions, loops, and connectors and arrange the flowchart in a way that reflects the organization and sequence of the code. **The flowcharts and the code needs to be linked on the generated interface.**

5. An online AI bird distribution system

Development of an online system to trace the distribution of birds. It encourages users to report birds around the place they live. Each report includes the location, the time, the name, and the images and a simple description of these birds. In addition, the system can automatically correct/remind users if users provide an incorrect name and suggest additional information such as their features and behaviors. One important function of this online system is it can dynamically display and compare the migration patterns of different types of birds. The AI component/chatbox can be used to answer the user questions regarding the collected data. The student teams are required to input sufficient data into their database for their demonstrations.

6. Analyzing cyber threats from log data (sample data will be provided):.

In this project, you are required to look into large network traffic data to detect communications that do not following the hand shaking mechanism of the TCP protocol. In particular, you need to list all of the potential issues found on the log data and potentially used for future predictions of suspicious communications. A visualization window allows users to look at the traffic data on a user selected range. The AI component/chatbox can be used to answer the user questions regarding the data. The student teams are required to link the visualization and the AI chatbox to support user understandings.



7. SitePro - Automation Software Solutions (www.sitepro.com)

SitePro projects 2026 are uploaded in a separate file

8. Toyota projects (will be presented in class on September 1).

<https://texastech.zoom.us/j/5720071948?pwd=S1lMQmFRM0VuZkFBOWFiM3BYK3phZz09>

9. Some more projects from TTU startups will be available soon